CHAPTER V:

LESSON 2

LEARNING OUTCOME 1:

SELECT PACKAGING MATERIALS

Objectives

At the end of the lesson, you are expected to:

2.1.1. Give the meaning of packaging, its importance and functions; and

2.1.2. Select appropriate packaging materials

Food packaging - is packaging of food which requires protection, tampering resistance, and special physical, chemical, or biological needs. It also shows in the product label any nutrition information on the food being consumed.

The main objective of packaging is to keep the food in good condition until it is sold and consumed, and to encourage customers to purchase the product. Correct packaging is essential to achieve both these objectives.

Packaging should provide the correct environmental conditions for food starting from the time food is packed through to its consumption. A good package should therefore perform the following functions:

Provide a barrier against dirt and other contaminants thus keeping the product clean

Prevent losses. For example, packages should be securely closed to prevent leakage.

Protect food against physical and chemical damage. For example the harmful effects of air, light, insects, and rodents. Each product will have its own need.

Package design should provide protection and convenience in handling and transport during distribution and marketing

Help the customers to identify the food and instruct them how to use it correctly

- Persuade the consumer to purchase the food
- Cluster or group together small items in one package for efficiency. Powders and granular materials need containment
- Marketing - The packaging and labels can be used by marketers to encourage potential buyers to purchase the product.
- Correct packaging prevents any wastage (such as leakage or deterioration) which may occur during transportation and distribution.

Types of Packaging Materials

In many developing countries the most commonly used food packaging materials include:

- leaves
- vegetable fibers
- wood
- papers, newsprint
- earthenware
- glass
- plastics
- metals

Leaves

Banana leaves are often used for wrapping certain types of food (e.g., suman). Corn husk is used to wrap corn paste or unrefined block sugar, and cooked foods of all types are wrapped in leaves. They do not however protect the food against moisture, oxygen, odors or microorganisms, and therefore, not suitable for long-term storage.

Vegetable fibers

These include bamboo, banana, coconut, and cotton fibres. These natural materials are converted into yarn, string or cord which will form the packaging material. These materials are very flexible, have some resistance to tearing, and are lightweight for handling and transportation. Being of vegetable origin, all of these materials are biodegradable and to some extent re-usable as with leaves, vegetable fibres do not provide protection to food which has a long shelf-life since they offer no protection against moisture pick-up, micro organisms, or insects and rodents.

Vegetable fiber basket Wood

Wooden shipping containers have traditionally been used for a wide range of solid and liquid foods including fruits, vegetables, tea and beer. Wood offers good protection, good stacking characteristics and strength. Plastic containers, however, have a lower cost and have largely replaced wood in many applications. The use of wood continues for some wines and spirits because the transfer of flavor compounds from the wooden barrels improves the quality of the product.

Paper

Paper is an inexpensive packaging material. It is however highly absorptive, fairly easily torn, and offers no barrier to water or gases. The degree of paper re-use will depend on its former use, and therefore paper that is dirty or stained should be rejected. Newsprint should be used only as an outer wrapper and not be allowed to come into direct contact with food, as the ink used is toxic.

Earthenware

Earthenware pots are used worldwide for storing liquids and solid foods such as curd, yoghurt, beer, dried food, and honey. Corks, wooden lids, leaves, wax, plastic sheets, or combinations of these are used to seal the pots

Glass

Glass has many properties which make it a popular choice as a packaging material:

- Glass is able to withstand heat treatments such as pasteurization and sterilization.
- Does not react with food.
- Protects the food from crushing and bruising
- Resistant to moisture, gases, odors and microorganisms
- Re-usable, re-sealable and recyclable
- Transparent, allowing products to be displayed. Coloured glass may be used either to protect the food from light or to attract customers

Disadvantages of using glass as packaging materials

- Glass is heavier than many other packaging materials and this may lead to higher transport costs
- It is easy to fracture, scratch and break if heated or cooled too quickly
- Potentially serious hazards may arise from glass cracks or fragments in the food.

Preparation of glass containers

- Inspection
- Washing.
- Rinsing.
- Sterilization.

- Sealing and capping
- Cooling

Plastics

The use of various plastics for containing and wrapping food depends on what is available. Plastics are extremely useful as they can be made in either soft or hard forms, as sheets or containers, and with different thickness, light resistance, and flexibility. The filling and sealing of plastic containers is similar to glass containers.

Flexible films are the most common form of plastic. Generally, flexible films have the following properties:

- Cost is relatively low.
- Good barrier properties against moisture and gases.

- Heat sealable to prevent leakage of contents.
- Have wet and dry strength.
- Easy to handle and convenient for the manufacturer, retailer, and consumer.
- Little weight to the product.
- Fit closely to the shape of the product, thereby wasting little space during storage and distribution.

Metal

Metal cans have a number of advantages over other types of containers:

- Metal cans provide total protection of the contents.
- Metal cans are tamperproof.
- Metal cans are convenient for presentation.

LEARNING OUTCOME 2:

PACKAGE FOOD ITEMS

Objectives

At the end of the lesson, you are expected to:

2.2.1. Package food items in compliance with Occupational Health and Safety Procedures;
2.2.2. Adopt appropriate packaging method according to enterprise standards; and
2.2.3. Label foods according to industry standards.

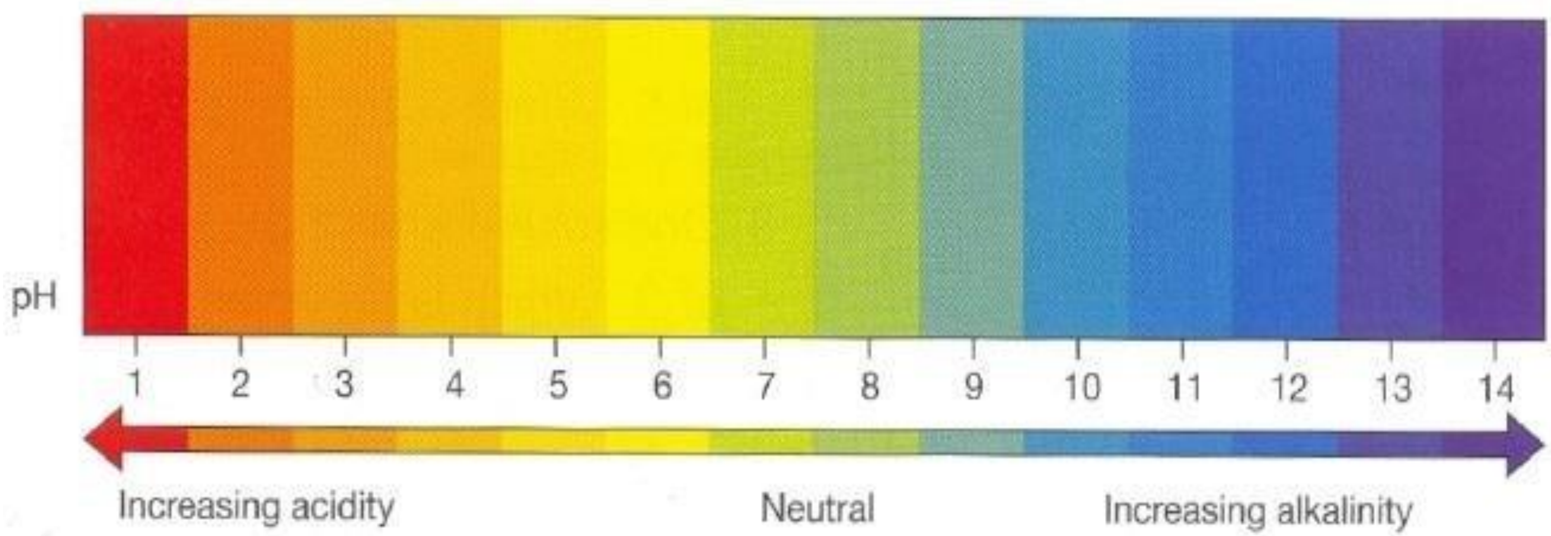
Food Safety on Storing and Transporting Foods

Food Packaging has been defined by Paine (1962) as the ―art science and technology of preparing goods for transport and sale‖. More specifically, it is a way of making sure that a product reaches the end user in good condition at the least cost to the producer. In effect, a packaging material provides the means of transporting a product from one place to another with maximum protection at the least cost

Food is packed in terms of quality, shelf life, microbial condition and portion control. Spoilage of food is caused by poor packaging of food. Below are some other factors in the spoilage of food:

a. **PH of food-** This simply indicates the inverse amount of hydrogen ion available in the food system. This is oftentimes associated with acidity of food. Thus, foods with high amounts of hydrogen ion have low pH and this is considered to have acidic taste. For instance, green mangoes have generally lower pH (Ph below 4.0) and therefore have high acidity.

b. **Moisture content** – This is related to the physical state of the food itself. Products with very high moisture could be those in liquid form, while those with very low free moisture could be dried or frozen. This product component is very important relative to food spoilage. The higher the moisture content, the greater the chances for microbial growth and chemical changes.



c. **Amount and nature of fat content** – The chemical processes also accelerate breakdown of fats on food. Thus, products with high fat content like oils, butter, soft cheese, fried foods and the like tend to spoil fast when inadequately packaged. Exposure to the atmosphere causes rapid oxidation breaking down the fat into free fatty acids in food. The faster the breakdown, the greater the chances of development of rancidity.

d. **Enzyme system** – It is a chemical processes like fermentation and hydrolysis which occur in high moisture food, especially in the presence of oxygen of ambient tropical temperature. When foods undergo these processes, they change in texture, flavor, odor and color. Under these conditions, the product may already be considered spoiled.

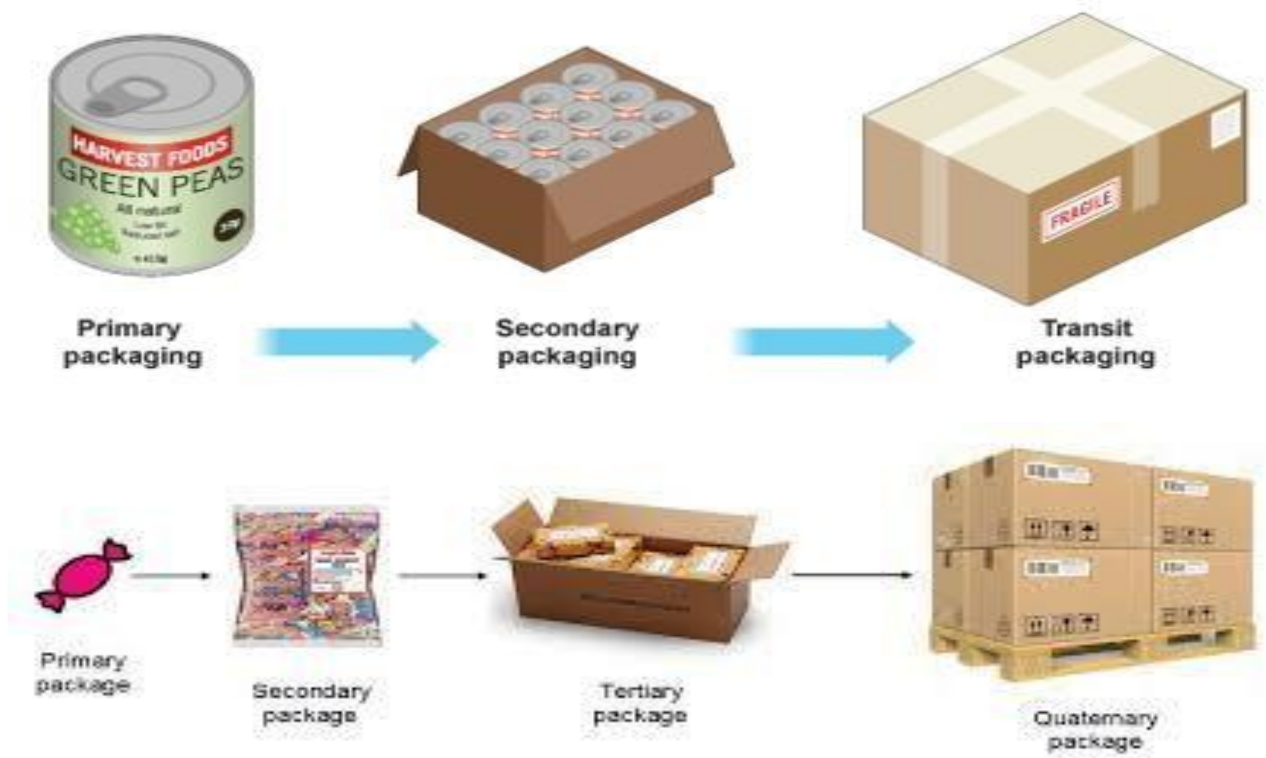
e. **Initial Microbial load** – This means that micro-organisms thrive best in high moisture foods like molds, yeast, and bacteria they grow faster in food with the high moisture content.

Bacteria These are the most important microorganisms to the food processor. Most are harmless, many are highly beneficial, some indicate the probable presence of filth, disease organisms, spoilage and a few cause disease.

Yeasts and Molds Yeasts are oval-shaped and slightly larger than bacteria. They reproduce most often by budding. They grow on most foods, on equipment, and building surfaces where there are small amounts of nutrient and moisture.

Viruses These are the smallest and simplest microorganisms. Unlike bacteria, yeasts, and molds, viruses are incapable of reproducing independently. However, bacteria find conditions of low pH, moisture, or temperature and high salt or sugar unfavorable. In such environments, yeasts or molds predominate. Thus, they can be a problem in dry foods, salted fish, bread, pickles, fruits, jams, jellies, and similar commodities.

CLASSIFICATION OF PACKAGING ACCORDING TO USE:



***As a primary package* –** This type of package is meant to directly contain the product. Hence, it gets in direct contact with the goods.

***As a secondary package* –** This is utilized to contain a specified number of unit packs. Thus, it may contain a dozen tetra packs or 2 dozens of tin cans or a gross of candies and so on. Its major function is to allow for the unit packs to be carried in bulk.

***As a tertiary package* –** When transporting in bulk, the secondary package may have to be packed again for greater protection and for bulk transfer. Use of tertiary package is normally for bulk transport or storage in large warehouses.

Occupational and health safety procedures in packaging foods

Steps on how to package meat before freezing.

1. Divide your meat into your set serving sizes
2. Get a plastic zip bag big enough to hold the portion size plus a little extra room for the meat to expand from the freezing process. 108
3. Put the meat into the bag, then flatten.
4. Squeeze as much of the air as you can get out, then close the zip.
5. Date the bag so you know how long it's been in your freezer

Methods of Food Packaging

Home Canned Foods - one of the oldest and most common methods of food packaging in homes is the use of home canning. Fruits and vegetables are placed in glass jars and sealed in the jars by heating the jars and then placing a rubber stopped jar top on the jar. The seals also need to be airtight to prevent the growth of bacteria.

Freezing and chilling food - another common method of packaging food is freezing and chilling .Freezing can be done with a variety of methods. Most often, it is vegetables that are frozen, although berries and other fruits can also lend themselves to being frozen.

Canned foods - canning foods as a method of food processing have been around, foods that are canned commercially are cooked prior to being placed in the can in order to prevent E. coli contamination. Canned foods come in a wide variety, ranging from meat to vegetables to fruit.

Foil Packaging - one of the innovative methods of commercial food packaging is foil wrapping. Foil wraps are often pouches that are filled and then the bottom and top of the pouch is sealed with a heat seal similar to those used with commercial frozen packaging. Foil packaging allows the foods to be sealed in the package without losing any residual moisture that may still be in the food. The best foods to package in this manner are usually dried fruits, baked goods or grain products.

What must appear on the label?

The following must appear on the label:

- Name under which the product is sold
- List of ingredients
- Quantity of certain ingredients
- Net quantity
- Date of minimum durability
- Any special storage instructions or conditions of use
- Name or business name and address of the manufacturer or packager, or of a seller

- Place of origin of the foodstuff if its absence might mislead the consumer to a material degree
- Instructions for use where necessary
- Beverages with more than 1.2% alcohol by volume must be declared